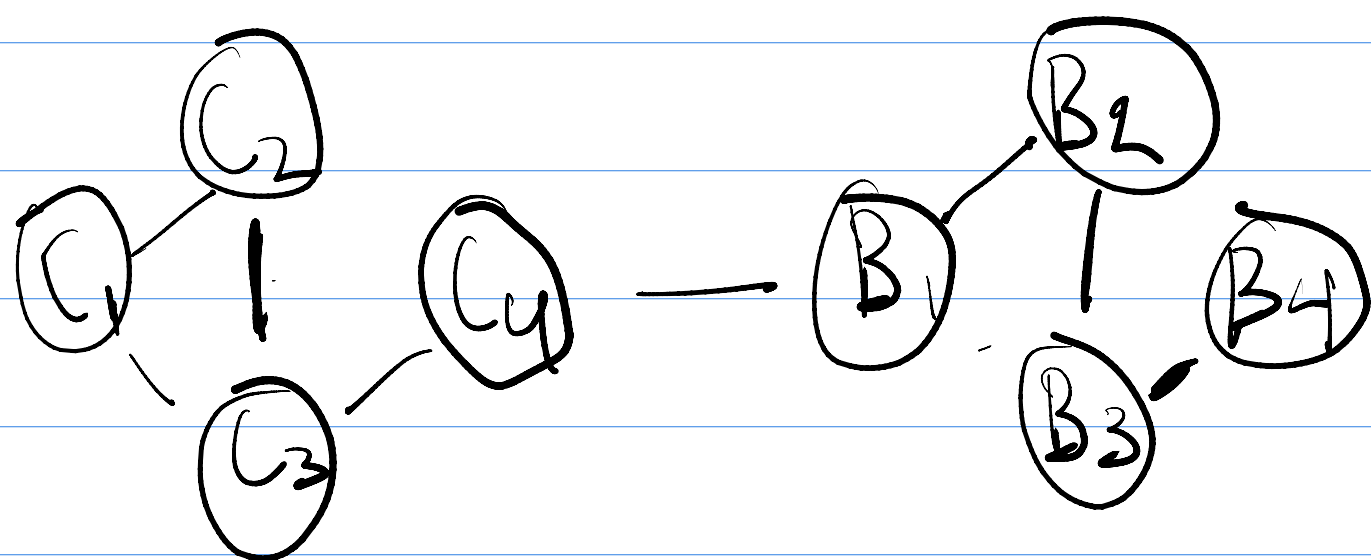


Networks - ex 06

6A1

Homophily

tendency for ties to form between similar nodes



① Edges within and across departments

CS = 4

Business = 3

Across =

② $E = 1, I = 7, E-I = (1-7) / (1+7) = -0.75$

③ in this case because it's strongly negative $\rightarrow$ strong homophily which means students from the same faculty befriends each other more

④ Random baseline = two groups of 4. expected cross group fraction $1 - \frac{\binom{4}{2} + \binom{4}{2}}{\frac{8}{2}} = 1 - \frac{12}{20} = 0.4$ so 57% of edges should be cross department under randomness. here's it's only 12.5% the $E-I$ index under random mixing would be $+0.14$

6A2

study > snippets > network-ex06 > 6.A.2.py > ...

```
1 import networkx as nx
2 import numpy as np
3
4 G = nx.karate_club_graph()
5
6 #club attributes as the group label
7 faction = nx.get_node_attributes(G, 'club')
8
9 #count within faction and cross faction edges
10 within_edges = 0
11 cross_edges = 0
12 for u, v in G.edges():
13     if faction[u] == faction[v]:
14         within_edges += 1
15     else:
16         cross_edges += 1
17
18 print(f"Within-faction edges: {within_edges}")
19 print(f"Cross-faction edges: {cross_edges}")
20
21 #compute E-I index for faction
22 E_I_index = (cross_edges - within_edges) / (cross_edges + within_edges)
23 print(f"E-I Index: {E_I_index:.3f}")
24
25 #Compute the expected cross-faction edge fraction under random mixing
26 n = len(G.nodes())
27 n1 = list(faction.values()).count('Mr. Hi')
28 n2 = n - n1
```

```

28 expected_cross_fraction = (2 * n1 * n2) / (n * (n - 1))
29
30 print(f"Expected cross-fraction fraction: {expected_cross_fraction:.3f}")
31

```

```

• > python3 6.A.2.py
Within-faction edges: 67
Cross-faction edges: 11
E-I Index: -0.718
Expected cross-fraction fraction: 0.515

```

6B.10 Projection: $S_1 - S_2$ (via C_1), $S_1 - S_3$ (via C_2), $S_3 - S_4$ (via C_3)

- ① Closures: Triadic: (S_2, S_1, S_3) is an open triad, predicts $S_2 - S_3$
 - Focal: S_1 and S_3 both attend C_2 , the shared course is a focal context
 - Membership: S_4 and S_3 r friend who attends C_2 , predicts S_4 may join C_2

② after S_2 joins C_2 : S_2 now shares C_2 with S_1 and S_3 forming $S_2 - S_3$ Focal closure
This also closes the triadic closure identified above

- ③ Info lost: - which course generates each tie
 - the course nodes themselves
 - tie-strength weights (multiple shared courses indicate stronger ties)

6B2

```

study > snippets > network-ex06 > 6.B.2.py > ...
1 import networkx as nx
2
3 #affiliation attributes as the group label S1,S2,S3,S4 and C1,C2,C3 enrollment S1-C1, S1-C2, S2-C1, S3-C2, S3-C3, S4-C3
4 G = nx.Graph()
5 students = ['S1', 'S2', 'S3', 'S4']
6 courses = ['C1', 'C2', 'C3']
7 G.add_nodes_from(students, bipartite=0)
8 G.add_nodes_from(courses, bipartite=1)
9
10 G.add_edges_from([('S1', 'C1'), ('S1', 'C2'), ('S2', 'C1'), ('S3', 'C2'), ('S3', 'C3'), ('S4', 'C3')])
11
12 #weighted projection
13 from networkx.algorithms import bipartite
14 student_projected = bipartite.weighted_projected_graph(G, students)
15 print(student_projected.edges(data=True))

```

Expected cross-fraction fraction: 0.515

```

• > python3 6.B.2.py
[('S1', 'S3', {'weight': 1}), ('S1', 'S2', {'weight': 1}), ('S3', 'S4', {'weight': 1})]

```

6C1 ① Socialization, the shift happens after friendship. Confirming requires longitudinal data on preferences before and after friendship formation.

② Confounding, Both uses python because of workplace; Friendship arose from co-presence
Comparing prehire language use would help isolate the confound

③ Socialization with confounding, Proximity (dorm) is the confound; Shared experiences create the convergent preference. Comparing same floor vs. different-floor pairs partially isolates proximity from pref convergence.

④ Socialization, preference changes after socialization.

6C2 ① Cross-sectional: group membership and current GPA. You'd find that group members have more similar GPA's than non-members

② GPA before the group forms. record group formation, measure GPA after group formation

③ Randomly assign students to group, eliminating selection. Any convergence is socialization

④ Selection: groups already shows above-average similarity

Socialization: post formation, similarity increases further

- 6D1
- ① $K=1$, No strong segregation. With only 1 of K same-type neighbour needed, almost everyone would be likely to be satisfied regardless of composition.
 - ② $K=5$, Strong segregation. Agents need a majority of neighbours to be the same type. This would cascade and make clusters.
 - ③ 30-40% same type neighbours but the dynamics amplify them.
 - ④ Schelling's model is spatial homophily. The "network" is the spatial grid itself: edges connect adjacent cells. The model shows that homophilic preferences, even weak ones, produce emergent structural segregation through feedback dynamics.

6D2.